

Supplementary material

S1. Data provenance

This retrospective secondary analysis used the BigP3BCI version 1.0.0 public archive. The downloaded archive had SHA256 digest eea294aa34e9ed11e5a25d07e30aee9df8b2d467a8309e2c38405a289afcd72f. Before any signal was processed, the ingestion pipeline checked this archive digest, read the distributor checksum manifest, selected non-AppleDouble European Data Format files, and checked every selected file against its manifest digest. The source archive and cache are not redistributed in this package.

All 20 documented source studies were examined. Every study supplied the 16 shared EEG channels sampled at 256 Hz together with the StimulusBegin, StimulusType and PhaseInSequence event channels, so no study was excluded for montage or event-channel incompatibility. Study-scoped participant identifiers were retained to prevent accidental cross-study linkage; they do not identify unique people across studies.

S2. Predictor and outcome reconstruction

The predictor used Train-phase files only. A calibration event was a rising StimulusBegin transition during phase 2, labelled by StimulusType. The 16 shared channels were filtered from 0.5 to 30 Hz with a fourth-order zero-phase Butterworth filter, epoched from 200 ms before to 800 ms after the event, baseline corrected to the prestimulus interval, and rejected at an absolute amplitude above 150 microvolts. A session entered the analysis only when at least 10 target and 40 non-target epochs survived rejection.

Epochs were decimated by taking every fourth sample. At 256 Hz this leaves 64 samples per channel and 1,024 features, and an effective sampling rate of 64 Hz. The resulting Nyquist frequency of 32

Hz stays above the 30 Hz upper edge of the bandpass, so the retained band is not folded onto lower frequencies, and the factor was chosen for that reason. Extraction rejects any file sampled below 240 Hz, the rate at which a factor of four would begin to fold the passband. The bandpass is a fourth-order zero-phase Butterworth rather than a brick wall, so attenuated content above 32 Hz does still fold onto 24 to 32 Hz; the filter response is 8.9 dB down at 32 Hz, that shoulder carries about 1.2% of output power for a white input, and mains at 60 Hz arrives about 60 dB down.

The number of cross-validation folds was the smaller of five and the number of Train files in the session. Fold count therefore differs between sessions and between studies, so the sampling precision of the predictor is not identical across the cohorts whose transportability is being compared.

For each Test-phase transition to phase 3, the intended character was the final nonzero CurrentTarget in the directly preceding phase-2 interval and the selected character was the modal nonzero SelectedTarget during phase 3. A trial was eligible when feedback was displayed, both values could be recovered, and FakeFeedback did not override the selection. Correctness required an exact character match.

Table S1. Feedback-phase reconstruction by source study. Eligible counts are at the feedback-phase level. Records additionally require a usable calibration feature set for the participant-session, which is why 19,611 selections enter the analysis rather than 19,688. The 77-selection difference is two participant-session-condition groups, both in Study B and both from participant B_01: sessions SE002 and SE003 were archived with 10 Test-phase recordings each and no Train-phase recording at all, so those sessions have no calibration epochs and no calibration score, and their 38 and 39 eligible selections cannot be estimated by a model whose predictor is that score. No other cohort loses a selection between the two totals. One further session, in Study C, was excluded at feature extraction because a single calibration file cannot support grouped

cross-validation, but Study C contributes no eligible online outcome in any case, so that exclusion does not move either total.

Source study	Reconstructed	Eligible	Dominant exclusion reason
Study A	1,404	1,404	none
Study B	858	858	none
Study C	5,680	0	artificial feedback override
Study D	1,230	1,230	none
Study E	240	240	none
Study F	1,079	1,067	feedback not displayed
Study G	1,198	1,198	none
Study H	1,926	1,926	none
Study I	948	948	none
Study J	1,812	1,812	none
Study K	480	480	none
Study L	990	990	none
Study M	1,260	1,260	none
Study N	480	480	none
Study O	1,202	1,187	feedback not displayed
Study P	2,263	0	intended character not recoverable
Study Q	3,888	1,944	feedback not displayed
Study R	1,440	1,440	none
Study S1	360	360	none
Study S2	864	864	none
Total	29,602	19,688	

S3. Model specification

The development model was a logistic regression of character-level correctness on the standardised calibration score, fitted on character-expanded records from the development studies, with standardisation using development-study means and standard deviations only. For a session with calibration score s , the estimated accuracy is the inverse logit of $a + b (s - m) / d$, where m and d are the development mean and standard deviation of the score. The four numbers are given for all 18 folds in Table S7, with a worked recomputation.

Expanding each record into one row per character weights a session by the number of characters it supplied, so the quantity estimated is the accuracy of a randomly chosen character selection. That choice is stated rather than left implicit, and the reported error does not rest on it. Repeating the whole leave-one-cohort-out evaluation with each session weighted equally, and again with each participant weighted equally, gave a pooled mean absolute error of 0.096 and 0.097 against 0.098 for the character weighting. Scored on its own scale each of the three fits had a calibration slope near one, at 0.966, 0.982 and 0.977; those three slopes are on three different scales by construction, so each is comparable to one and not to the others. The three rows are written to `output/expanded/estimand_comparison.csv` by `scripts/09_run_estimand.py`.

S4. Uncertainty

Two uncertainty statements are reported and are not interchangeable.

The participant-cluster bootstrap used 2,000 deterministic replicates. In each replicate, development participant clusters were resampled within source study, the model was refitted, and withheld participant clusters were resampled. The resulting intervals describe uncertainty conditional on the observed set of source studies, because the same studies appear in every replicate by construction.

The study-level summary treats the source study as the unit of replication. The withheld-cohort estimates were summarised by their mean and between-study standard deviation, with a t-distributed interval for the mean on $k - 1$ degrees of freedom and a prediction interval for an unrepresented cohort computed as the mean plus or minus t times the between-study standard deviation times the square root of one plus one over k .

Table S2. Study-level summaries. The calibration intercept and slope rows are the uncorrected summary of the 18 cohort estimates, which counts each cohort's sampling error as though it were between-cohort variation. For those two quantities the main text reports the random-effects estimate τ instead, and the intervals here should not be read in its place.

Quantity	Mean across cohorts	Between-cohort SD	95% interval for the mean	95% interval for an unrepresented cohort
Mean absolute error	0.101	0.047	0.077 to 0.124	-0.002 to 0.203
Brier skill score	0.174	0.217	0.066 to 0.282	-0.296 to 0.644
Character-weighted AUC	0.713	0.102	0.662 to 0.764	0.491 to 0.935
Calibration intercept	-0.008	1.175	-0.592 to 0.576	-2.555 to 2.539
Calibration slope	1.100	0.560	0.822 to 1.379	-0.113 to 2.313

S5. Association at three levels

Table S3. Association between calibration-derived decoder discriminability and observed accuracy, by level and by cohort. Pearson intervals are two-sided 95% intervals on the Fisher z scale. The two preceding-session rows order a participant's sessions by the lexical order of their identifiers, which the archive assigns sequentially within a participant; the de-identified timestamps cannot confirm that ordering, so those two rows rest on the assumption that identifier

order matches recording order.

By level.

Level	n	Pearson r	95% CI	Spearman rho
Sessions, pooled	410	0.716	0.665 to 0.760	0.718
Sessions, centred within cohort	410	0.677	0.621 to 0.726	0.646
Participants	271	0.714	0.650 to 0.768	0.756
Preceding session to later session	139 pairs	0.513	0.379 to 0.626	0.462
Same session, reference for the row above	139 pairs	0.728	0.639 to 0.798	0.651

By cohort. ALS cohorts are listed first. The unit is the participant-session, so n is smaller than the record count in Table S2.

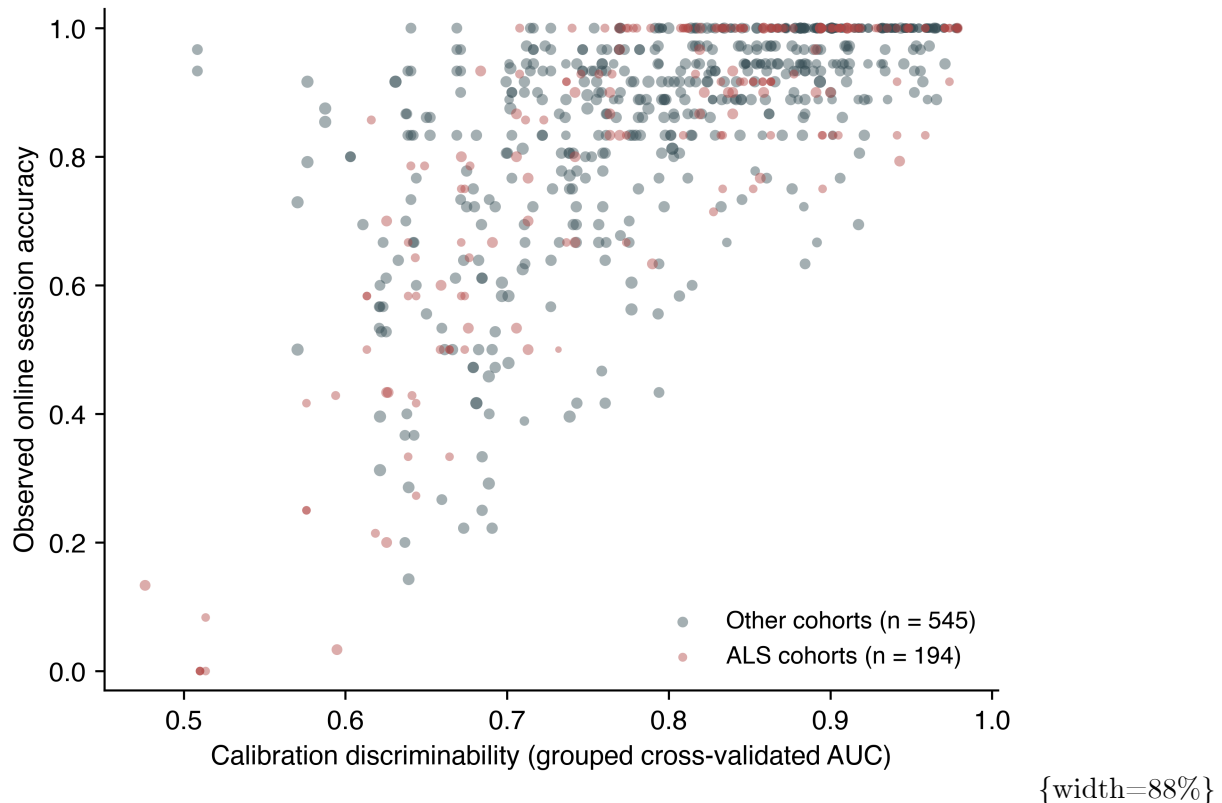
Cohort	Sessions	Pearson r	95% CI
Study B	56	0.678	0.505 to 0.798
Study F	30	0.924	0.846 to 0.964
Study L	11	0.922	0.719 to 0.980
Study N	16	0.823	0.553 to 0.937
Study A	13	0.928	0.771 to 0.979
Study D	17	0.610	0.183 to 0.843
Study E	8	0.683	-0.042 to 0.937
Study G	20	0.757	0.473 to 0.899
Study H	16	0.190	-0.338 to 0.627
Study I	13	0.513	-0.053 to 0.830
Study J	20	0.359	-0.099 to 0.692
Study K	8	0.401	-0.424 to 0.862
Study M	21	0.660	0.320 to 0.850

Cohort	Sessions	Pearson r	95% CI
Study O	34	0.364	0.030 to 0.625
Study Q	53	0.773	0.636 to 0.863
Study R	40	0.425	0.131 to 0.651
Study S1	10	0.316	-0.392 to 0.789
Study S2	24	0.609	0.273 to 0.813

The point estimate was positive in all 18 contributing cohorts. Within-cohort Pearson r ranged from 0.190 to 0.928 with a median of 0.635, and the 95% interval included zero in 6 of the 18: Study E, Study H, Study I, Study J, Study K and Study S1. All six carry 20 sessions or fewer, so the imprecision is largely a matter of cohort size rather than of a different relationship.

Session accuracy clustered within participant with an intraclass correlation of 0.412 across 410 sessions in 271 participants, giving approximately 338 effective independent sessions. The predictor is constant within a session, so the 739 session-condition records carry 410 distinct predictor values.

Figure S1. Calibration-derived decoder discriminability against observed online accuracy, by cohort type. Each point is one participant-session-condition record, sized by the number of eligible selections. No line is fitted to either group. Cohort type does not vary between the records of a cohort, so the comparison of the ALS cohorts against the rest is made on the 18 cohort-specific slopes, reported in the main text and drawn in Figure 1; fitting it across these records would treat a study-level attribute as though it had been measured once per record. The band of points at 1.0 is the ceiling described in the main text.



S6. Sensitivity analyses

Table S4. Sensitivity analyses. The four analyses named in the Methods, restriction to cohorts with meaningful outcome variance, exclusion of records with high artifact rejection, exclusion of small-denominator records, and leave-two-studies-out development, were listed before any of them was run on the widened cohort; none was registered. The primary row and the two cohort-subset rows are shown beside them for comparison. Each analysis re-runs the complete withheld-cohort procedure on the retained data. In the last row every pair of cohorts is withheld together, so development runs on 16 cohorts rather than the 17 of the primary analysis; each cohort appears in 17 of the 153 pairs and its 17 estimates are averaged into a single value before the same across-cohort pooling is applied, because a pair shares a cohort with 32 other pairs and the pairs are not independent units. Every row therefore summarises 18 cohort-level values, or fewer where the

analysis drops cohorts, and the columns carry the same meaning throughout.

Analysis	Cohorts	Mean absolute error	Between- cohort SD	Interval for an unrepresented cohort	Calibration- slope SD
Primary, all contributing cohorts	18	0.101	0.047	-0.002 to 0.203	0.560
Cohorts with outcome SD at least 0.10	12	0.124	0.042	0.029 to 0.220	0.601
Records with artifact rejection at most 20%	18	0.087	0.035	0.011 to 0.164	0.465
Records with at least 10 eligible selections	18	0.100	0.047	-0.002 to 0.203	0.560
Cohorts without a documented ALS population	14	0.101	0.052	-0.015 to 0.216	0.663
ALS cohorts, originally targeted primary subgroup	4	0.093	0.016	0.035 to 0.151	0.223
Leave-two-studies-out development, 153 splits	18	0.101	0.047	-0.002 to 0.203	0.560

Artifact rejection had a median of 0.002 across records, but 110 of 739 records exceeded 20% and the maximum was 0.97. The analysis restricted to records at or below 20% rejection is the only one in which both estimation error and calibration-slope variability improved.

Observed session-condition accuracy was at 100% in 228 of 739 records (30.9%). Three cohorts had mean accuracy at or above 0.96, where there is little variation to estimate.

The leave-two-studies-out row is indistinguishable from the primary row at the precision printed above, and that is the result rather than a rounding artefact. At four decimal places the mean

absolute error is 0.1007 against 0.1006, its between-cohort standard deviation 0.0473 against 0.0472, and the uncorrected calibration-slope standard deviation 0.5599 against 0.5596. Removing one cohort from a development set of 17 neither raises the estimation error nor widens the between-cohort spread, so the transportability failure is not a consequence of the development set being too small at this scale. It says nothing about development sets much smaller than 16, which this design cannot examine while still leaving enough cohorts to withhold.

S7. The two no-predictor benchmarks, per cohort

Skill in the main text is computed against the development-mean benchmark, which estimates every withheld record at the development-set mean accuracy. A second benchmark estimates every withheld record at that cohort's own mean, which no deployment would know. The two are different comparisons and give different counts of cohorts with negative skill, so both are given here per cohort.

Pooled, the own-mean benchmark is the harder of the two, at 0.123 against 0.146. That ordering does not hold cohort by cohort. Mean absolute error is minimised by the median rather than the mean, so a cohort's own mean is not guaranteed to beat any other constant, and it is in fact the easier target in Study A, Study F, Study K, Study L, Study M, Study N and Study Q.

Skill against the development mean is negative in one cohort, Study H. Skill against the cohort's own mean is negative in six: Study E, Study H, Study J, Study R, Study S1 and Study S2. Study S1's value is an artefact of a near-zero denominator rather than a comparable failure, because its own-mean benchmark errs by 0.005; it is reported for completeness and should not be read on the same scale as the others.

Table S5. Model error against both no-predictor benchmarks, by withheld cohort.

ALS cohorts are listed first. Skill is one minus the ratio of the model's mean absolute error to that benchmark's.

Cohort	Mean observed accuracy	Model MAE	Development-mean benchmark MAE	Skill vs development mean	Own-mean benchmark MAE	Skill vs own mean
Study B	0.899	0.108	0.158	0.317	0.132	0.185
Study F	0.778	0.101	0.203	0.502	0.221	0.543
Study L	0.839	0.084	0.137	0.383	0.140	0.397
Study N	0.694	0.124	0.226	0.448	0.228	0.454
Study A	0.786	0.126	0.173	0.271	0.178	0.291
Study D	0.896	0.056	0.092	0.394	0.071	0.211
Study E	0.921	0.063	0.079	0.205	0.049	-0.286
Study G	0.886	0.066	0.113	0.417	0.099	0.336
Study H	0.877	0.153	0.121	-0.268	0.109	-0.407
Study I	0.658	0.172	0.226	0.236	0.208	0.171
Study J	0.708	0.186	0.187	0.003	0.185	-0.008
Study K	0.750	0.173	0.193	0.104	0.206	0.160

	Mean observed	Model	Development-mean	Skill vs	Own-mean	Skill vs own
Cohort	accuracy	MAE	benchmark MAE	development mean	benchmark MAE	mean
Study	0.823	0.121	0.134	0.093	0.138	0.118
M						
Study	0.884	0.067	0.081	0.181	0.070	0.046
O						
Study	0.820	0.058	0.096	0.394	0.099	0.409
Q						
Study	0.963	0.059	0.131	0.547	0.047	-0.263
R						
Study	0.997	0.043	0.151	0.717	0.005	-7.090
S1						
Study	0.976	0.048	0.141	0.657	0.035	-0.363
S2						
Pooled	0.851	0.098	0.146	0.327	0.123	0.203

S8. Comparator predictors

Comparator scores were computed from the identical calibration epochs and evaluated under the identical withheld-cohort protocol: regularised linear discriminant analysis area under the curve, grouped cross-validated classification accuracy, mean target-minus-non-target amplitude at Pz between 250 and 500 ms, the same contrast averaged over six posterior channels, and the maximum posterior signed r-squared. An exploratory specification adds ALSFRS-R to the primary score.

Table S6. Withheld-cohort performance of every predictor named in the Methods.

Pooled values are computed over all withheld predictions, as in the main text. The last five columns summarise the cohort-level results and are the transportability quantities: the mean across cohorts

is the quantity Table S4 reports as mean absolute error and is what the interval is centred on, the interval is for a cohort not represented in the archive, and the calibration-slope spread is uncorrected, as in Table S4. Pooled MAE and the mean across cohorts are different quantities and the first is not the centre of the interval. Rows are ordered by pooled error within each role. **The last two rows rest on 3 cohorts and 138 records, not 18 and 739, so their apparently lower error is a property of that restricted set and is not comparable with the rows above.**

Between-cohort												
Predictor	Role	Cohort	Records	Pooled		Mean across cohorts	MAE	Interval for an unrepre-sented cohort	Calibration		Calibration-slope	
				Brier skill	AUCslope				slope	slope		
				MAE	SD	SD	range					
Calibration-derived decoder discriminability	Primary	18	739	0.098	0.110	0.748	0.967	0.101	0.047	-0.002 to 0.203	0.560	0.185 to 2.185
Shrinkage linear discriminant AUC	Comparator	18	739	0.100	0.105	0.745	0.960	0.102	0.046	0.001 to 0.202	0.562	0.112 to 2.077
Calibration classification accuracy	Comparator	18	739	0.106	0.105	0.734	0.930	0.107	0.047	0.006 to 0.208	1.104	0.051 to 4.018
Maximum posterior signed r-squared	Comparator	18	739	0.134	0.037	0.656	0.643	0.135	0.041	0.045 to 0.225	2.209	-0.467 to 7.297

Between-													
Mean cohort Interval for CalibrationCalibration-													
PooledBrier Pooled across MAE an unrepre- slope slope													
Predictor	Role	Cohort	Records	MAE	skill	AUC	slope	cohorts	MAE	SD	sented cohort	SD	range
Posterior	Comparator	18	739	0.148	0.003	0.517	0.275	0.149	0.040	0.061	to	2.459	-4.745 to
target-minus-										0.236			6.387
non-target													
amplitude													
Pz	Comparator	18	739	0.150	0.001	0.465	-	0.150	0.040	0.063	to	4.490	-8.235 to
target-minus-							0.238			0.238			9.007
non-target													
amplitude													
Primary	Exploratory	3	138	0.085	0.313	0.834	0.982	0.090	0.009	0.046	to	0.293	0.771 to
score plus										0.133			1.336
ALSFRS-R													
Primary	Exploratory	3	138	0.085	0.315	0.833	0.995	0.090	0.010	0.038	to	0.303	0.765 to
score, same	refer-									0.142			1.355
restricted	ence												
records													

No confidence intervals from the participant bootstrap are reported in this table. The primary predictor's intervals are given in the main text from 2,000 replicates, and rerunning that bootstrap at a smaller replicate count for a seven-specification sweep would put a second, slightly different interval for the same primary quantity into the same paper. The uncertainty reported instead is the between-cohort spread, which requires no resampling and is the quantity the transportability question turns on. The primary row reproduces the pooled values reported in the main text on every column shown here, which is the consistency check this table can offer.

The regularised linear discriminant computed from the identical features agrees with the primary score to within 0.007 on every summary column, the largest gap being the pooled calibration slope at 0.960 against 0.967 and the between-cohort spread of that slope at 0.562 against 0.560. The per-cohort slope range is the one column where they separate, 0.112 to 2.077 against 0.185 to 2.185. The primary result is therefore a property of how separable the calibration data are and not of the classifier family used to measure that. It carries no further implication, because the two classifiers read the same features and so cannot separate any question about how those features were built.

Scoring the same classifier by classification accuracy rather than by area under the curve costs little in pooled error, 0.106 against 0.098, but doubles the spread of the calibration slope, to 1.104 across a range of 0.051 to 4.018. A thresholded summary of the same calibration data transports worse than the ranking one.

The three summaries computed without fitting a classifier are much weaker. The maximum posterior signed r-squared reaches an area under the curve of 0.656, and neither amplitude contrast separates outcomes in a cohort it was not developed on, at 0.517 and 0.465 with Brier skill of 0.003 and 0.001. Their pooled calibration slopes of 0.275 and -0.238, and per-cohort slope ranges of -4.7 to 6.4 and -8.2 to 9.0, are what the calibration parameters look like when a predictor carries almost no transportable signal. They are reported so that the primary score's own spread is read against that scale rather than in isolation.

The exploratory ALSFRS-R specification is restricted to the 138 records in 3 cohorts that carry an observed value, so its numbers are not comparable with the rows above. Against the primary score run on those same 138 records it changes nothing: pooled error 0.085 against 0.085, area under the curve 0.834 against 0.833, Brier skill 0.313 against 0.315, and calibration-slope spread 0.293 against 0.303. The better appearance of both rows relative to the full archive belongs to the restriction

and not to ALSFRS-R. At 3 cohorts the interval for an unrepresented cohort rests on two degrees of freedom and carries no transportability claim.

S9. Reproducibility

Recomputing any estimate

Table S7. Fitted coefficients of every development fold. One row per withheld cohort. The estimate for a session with calibration score s in the fold that withheld a given cohort is the inverse logit of $a + b(s - m) / d$, using that row's four numbers. Coefficients are on the standardised scale, so b is the change in log odds per development standard deviation of the score.

Withheld cohort	Development records	Development selections	a	b	m	d
Study A	700	18,207	2.0672	0.9766	0.80553	0.10820
Study B	683	18,830	2.0030	0.9688	0.80476	0.10689
Study D	705	18,381	2.0155	0.9934	0.80354	0.10794
Study E	731	19,371	2.0209	0.9783	0.80449	0.10657
Study F	650	18,544	2.0281	0.8991	0.80508	0.10197
Study G	699	18,413	2.0127	0.9852	0.80218	0.10761
Study H	675	17,685	2.0362	1.0357	0.80886	0.10788
Study I	713	18,663	2.0823	0.9493	0.80817	0.10570
Study J	699	17,799	2.1463	1.0333	0.80942	0.10474
Study K	723	19,131	2.0624	1.0001	0.80719	0.10470
Study L	706	18,621	2.0219	0.9638	0.80551	0.10730
Study M	697	18,351	2.0492	0.9925	0.80866	0.10604
Study N	723	19,131	2.0458	0.9561	0.80577	0.10548
Study O	705	18,424	2.0442	1.0220	0.80138	0.10704
Study Q	685	17,667	2.0533	0.9883	0.80842	0.10811

Withheld cohort	Development records	Development selections	a	b	m	d
Study R	659	18,171	1.9028	0.9545	0.79471	0.10654
Study S1	719	19,251	1.9809	0.9553	0.80105	0.10529
Study S2	691	18,747	1.9400	0.9513	0.79803	0.10621

Worked example. The first record of the cohort withheld in the Study H fold has a calibration score of 0.710714. From that row, $m = 0.80886$ and $d = 0.10788$, so the standardised score is $(0.710714 - 0.80886) / 0.10788 = -0.9098$. With $a = 2.0362$ and $b = 1.0357$ the log odds are $2.0362 + 1.0357 \times (-0.9098) = 1.0940$, and the inverse logit of that is 0.7491, which is the estimate the analysis records for that session-condition.

Recomputing all 739 held-out estimates from the printed values reproduces the frozen predictions to within 0.0001. The coefficients themselves are not refitted for this table; they are read off the fitted model at two scores, and every fold is checked by reconstructing its own predictions from the four numbers before the table is written.

Pipeline and outputs

The analysis pipeline executes archive validation, source metadata extraction, feedback-phase reconstruction, calibration feature extraction, withheld-cohort validation, the widened-design analyses, per-cohort calibration and heterogeneity, predictor precision, the estimand comparison, the protocol-moderator analysis, sensitivity analyses, the comparator-predictor sweep, the fold-coefficient export, and the rendering of every figure from those frozen outputs. Frozen outputs are written to `output/expanded/`: `study_inventory.csv`, `analysis_records.csv`, `external_validation_metrics.csv`, `external_validation_predictions.csv`, `random_effects_pooling.csv`, `als_subgroup_metrics.csv`, `transfer_to_als.csv`, `als_meta_regression.json`,

null_benchmark.csv, within_study_association.csv, participant_level_association.csv, across_session_association.csv, across_session_pairs.csv, session_clustering.json, cohort_calibration.csv, heterogeneity_summary.json, predictor_precision.csv, predictor_reliability.csv, estimand_comparison.csv, sensitivity_analyses.csv, comparator_metrics.csv, fold_coefficients.csv, protocol_covariates.csv and protocol_meta_regression.json.

The repository test suite contains 162 tests covering provenance, European Data Format parsing, event reconstruction, feature extraction, validation, the widened-design analyses, the promised sensitivity, comparator and fold-coefficient analyses, the session-ordering assumption behind the preceding-session analysis, and rendering.

S10. Transparency statement

The study involved no new data collection, participant contact, prospective enrolment, or intervention. It does not establish a diagnostic, prognostic, causal, or treatment effect. Character-level online selection accuracy is an operational endpoint and should not be presented as communication success, quality of life, or a clinical outcome. The source studies vary in protocol, and the archive does not permit cross-study person-level linkage. Four source studies carry a documented ALS population; the remaining cohorts are described as other cohorts because the documentation does not support a positive characterisation, and no participant-level clinical characteristics were available.